

Filled Subgrid Count II

Time limit: 1.00 s Memory limit: 512 MB

You are given a grid of letters. Your task is to calculate, for each letter, the number of *rectangular* subgrids whose each letter is the same.

Input

The first line has two integers n and k : the size of the grid and the number of letters. The letters are the first k uppercase letters.

After this, there are n lines that describe the grid. Each line has n letters.

Output

Print k lines: for each letter, the number of subgrids.

Constraints

- $1 \leq n \leq 3000$
- $1 \leq k \leq 26$

Example

Input	Output
5 3 ABBBC BBBBC BCAAA AAAAA AAAAA	64 24 4